

Hennekam Lymphangiectasia-Lymphedema Syndrome 2 (HKLLS2): Comprehensive Disease Characteristics Report

Disease: Hennekam lymphangiectasia-lymphedema syndrome 2 (HKLLS2) **OMIM:** #616006 · **MONDO:** MONDO:0014217 · **Orphanet:** ORPHA:2136 (Hennekam syndrome) · **ICD-10:** Q82.0 (hereditary lymphedema) **Causal gene:** *FAT4* (FAT atypical cadherin 4; HGNC:23109; 4q28.1) · **Inheritance:** Autosomal recessive **Category:** Mendelian

Summary

Hennekam lymphangiectasia-lymphedema syndrome 2 (HKLLS2) is an ultra-rare, autosomal recessive generalized lymphatic dysplasia caused by **biallelic loss-of-function variants in *FAT4***, an atypical Fat/Dachsous-family cadherin located at 4q28.1. It is the second genetically defined form of Hennekam syndrome — HKLLS1 is caused by *CCBE1* and HKLLS3 by *ADAMTS3* — and is **allelic to Van Maldergem syndrome (VMS)**, which arises from *FAT4* or *DCHS1* mutations and shares a facial gestalt and intellectual disability but generally lacks the lymphedema that defines Hennekam syndrome. The disease was molecularly delineated by Alders and colleagues in 2014 through homozygosity mapping and whole-exome sequencing of the original Hennekam family, in whom no *CCBE1* mutation had been found ([PMID: 24913602](#)).

Mechanistically, *FAT4* acts **cell-autonomously in lymphatic endothelial cells (LECs)** to establish planar cell polarity (PCP) in response to fluid flow, working with its binding partner Dachsous1 (*DCHS1*) to drive lymphatic valve and vessel morphogenesis. Loss of *FAT4* disrupts LEC polarization, causing defective lymphatic valves and dysfunctional lymphatic drainage; this converges on the same **VEGF-C/VEGFR3 prolymphangiogenic axis** in which *CCBE1* and *ADAMTS3* operate (by proteolytically activating pro-VEGF-C). *FAT4/DCHS1* signaling additionally regulates Hippo/YAP-dependent cortical neurogenesis, which explains the neurodevelopmental features. The downstream clinical consequences form a characteristic

tetrad: **congenital/generalized lymphedema, intestinal lymphangiectasia with protein-losing enteropathy (PLE), facial dysmorphism, and variable (mild-to-moderate) intellectual disability**, with additional multisystem involvement (chylous effusions, pericardial effusion, hearing loss, hypogammaglobulinemia).

There is **no curative therapy**. Management is supportive and multidisciplinary: complete decongestive therapy for lymphedema, high-protein/low-fat/medium-chain-triglyceride nutrition and albumin/immunoglobulin support for intestinal lymphangiectasia and PLE, drainage/management of effusions, and symptomatic care of dysmorphic, cognitive, and sensory features. Diagnosis is established by **molecular genetic testing** (exome/genome sequencing or a primary-lymphedema gene panel including *CCBE1*, *FAT4*, *ADAMTS3*) within the standardized primary lymphatic dysplasia diagnostic pathway, supported by imaging (lymphoscintigraphy, MR lymphangiography), intestinal biopsy showing lymphangiectasia, and laboratory evidence of hypoalbuminemia and hypogammaglobulinemia. Prevention relies on genetic counseling with the option of prenatal or preimplantation genetic testing in at-risk families.

Evidence base: this is an ultra-rare disorder (~50 Hennekam-syndrome cases reported worldwide; the *FAT4* subtype a minority). Nearly all information derives from **individual patient case reports/small cohorts (human clinical), model-organism** studies (mouse, *Drosophila*), and **in vitro** biochemistry — not from large aggregated registries or EHR datasets. Where data are absent, this is stated explicitly.

Key Findings

Finding 1 — HKLLS2 is caused by biallelic loss-of-function *FAT4* variants (autosomal recessive; OMIM #616006)

The genetic basis of HKLLS2 was established by Alders et al. (2014). Using homozygosity mapping combined with whole-exome sequencing in the original Hennekam family — in whom *CCBE1* mutations had been excluded — they identified a homozygous *FAT4* mutation, then confirmed the gene by targeted analysis of a *CCBE1*-negative cohort:

"We used homozygosity mapping and whole-exome sequencing in the original HS family with multiple affected individuals in whom no CCBE1 mutation had been detected, and identified a homozygous mutation in the *FAT4* gene. Subsequent targeted mutation analysis of *FAT4* in a cohort of 24 CCBE1 mutation-negative Hennekam syndrome patients identified homozygous or compound heterozygous mutations in four additional families."
— [PMID: 24913602](#)

The autosomal recessive inheritance and the OMIM identifier #616006 for the *FAT4*-associated form are corroborated in the ADAMTS3 delineation paper:

"It is an autosomal recessive condition caused by biallelic mutations in CCBE1 ... (HKLLS1; OMIM 235510) or *FAT4* (HKLLS2; OMIM 616006)." — [PMID: 30450763](#)

Reported *FAT4* variants are biallelic — homozygous in consanguineous families or compound heterozygous — and predominantly loss-of-function: missense, nonsense, frameshift, and splice-site changes. A 2026 report described a novel biallelic intron-14 splice variant (c.12479+3A>G) causing HKLLS2, with aberrant splicing confirmed by RT-PCR of patient fibroblasts ([PMID: 41992670](#)). *FAT4* is allelic to Van Maldergem syndrome, so the same gene produces distinct but overlapping phenotypes depending on variant and modifier context.

Finding 2 — *FAT4* controls LEC polarity and lymphatic valve morphogenesis, converging on the VEGF-C/VEGFR3 pathway shared by all Hennekam genes

FAT4 is the molecular linchpin of HKLLS2 pathophysiology. Betterman et al. (2020) demonstrated that *FAT4* functions within LECs to sense flow and establish polarity, and that it is a transcriptional target of the lymphatic master regulator GATA2:

"we demonstrate that *FAT4* functions in a lymphatic endothelial cell-autonomous manner to control cell polarity in response to flow and is required for lymphatic vessel morphogenesis throughout development" — [PMID: 32182215](#)

Pujol et al. (2017) showed that the Dachshous1–*Fat4* PCP module directs endothelial cell polarization specifically during **lymphatic valve morphogenesis**, and directly linked valve defects to the lymphedema seen in *FAT4*-Hennekam syndrome:

"Our data demonstrate that *Fat4* and *Dachsous1* are critical regulators of valve morphogenesis. This study highlights that valve defects may contribute to lymphedema in Hennekam syndrome caused by *Fat4* mutations." — [PMID: 28705793](#)

Crucially, all three principal Hennekam genes act in a shared prolymphangiogenic pathway. CCBE1 (a matrix protein) and ADAMTS3 (a protease) proteolytically cleave and activate pro-VEGF-C, enabling VEGFR3 signaling, while *FAT4* governs the endothelial polarity required to build functional vessels and valves:

"encoding a matrix protein and protease, respectively, that regulate activity of the key prolymphangiogenic VEGF-C/VEGFR3 signaling axis by facilitating the proteolytic cleavage and activation of VEGF-C. The fact that *FAT4*, *CCBE1*, and *ADAMTS3* mutations underlie Hennekam syndrome suggested that all 3 genes might function in a common pathway" — [PMID: 32182215](#)

The VEGF-C activation mechanism is further supported by biochemical work showing that an *ADAMTS3*–*CCBE1* complex is required to convert VEGF-C (but not VEGF-D) into an active VEGFR3 ligand ([PMID: 27159393](#)) and that the two domains of *CCBE1* promote VEGFR3 signaling by distinct mechanisms ([PMID: 28687807](#)). *PROX1* (master LEC identity factor) and VEGFR-3 form the developmental backbone of this system ([PMID: 29842991](#)).

Finding 3 — Core clinical phenotype: congenital lymphedema, intestinal lymphangiectasia/PLE, facial dysmorphism, and variable intellectual disability

Ivanovski, Alders, and Hennekam (2018) reviewed all reported *FAT4* patients (VMS n=11, HS n=40), delineating the shared and distinguishing features of the two allelic conditions:

"Both conditions are characterized by a typical facial gestalt and mild to moderate intellectual disability, but differ in the occurrence of neonatal hypotonia and feeding problems, hearing loss, tracheal anomalies, and osteopenia in VMS, and lymphedema in HS." — [PMID: 29681106](#)

Individual case reports document the full clinical spectrum: generalized/peripheral lymphedema, facial anomalies (hypertelorism, flat nasal bridge, flat facial profile), camptodactyly, hypotonia, and duodenal lymphangiectasia with hypoalbuminemia and hypogammaglobulinemia ([PMID: 25616299](#), [PMID: 41992670](#)). The intestinal lymphangiectasia (a form of Waldmann disease) drives protein-losing enteropathy as a downstream consequence:

"Primary intestinal lymphangiectasia (WD) is a consequence of HS, which ultimately results in PLE and worsening interstitial lymph buildup." — [PMID: 38028107](#)

Finding 4 — Model organisms recapitulate PCP, kidney, neural, and lymphatic phenotypes

Fat4-null mice and *Drosophila fat* mutants provide strong cross-species mechanistic support. In mouse (NCBI Taxon 10090; gene *Fat4*), Saburi et al. (2008) established the PCP role:

"Loss of *Fat4* disrupts oriented cell divisions and tubule elongation during kidney development, leading to cystic kidney disease. *Fat4* genetically interacts with the PCP genes *Vangl2* and *Fjx1* in cyst formation." — [PMID: 18604206](#)

Cappello et al. (2013) recapitulated the neurodevelopmental component and linked *FAT4*/*DCHS1* to Hippo/YAP signaling:

"Reducing the expression of *Dchs1* or *Fat4* within mouse embryonic neuroepithelium increased progenitor cell numbers and reduced their differentiation into neurons, resulting in the heterotopic accumulation of cells below the neuronal layers in the neocortex, reminiscent of the human phenotype. These effects were countered by concurrent knockdown of *Yap*" — [PMID: 24056717](#)

The lymphatic-specific phenotype is captured by *Fat4*-deficient mice with defective valve morphogenesis and endothelial polarity ([PMID: 28705793](#), [PMID: 32182215](#)). In *Drosophila* (Taxon 7227; ortholog *fat*), neuron-specific *fat* knockdown provides a model relevant to the intellectual-disability/neuronal features:

"Neuron-specific knockdown of fat shortened the life span and induced the defect in locomotive abilities of adult flies... Both VMS and HS show mental retardation and neuronal defects. We therefore consider that these two rare human diseases could possibly be caused by the defect in FAT4 function in neuronal cells." — [PMID: 28488382](#)

Finding 5 — Management is supportive; no curative therapy exists

There is no disease-modifying or curative treatment for HKLLS2. Lee et al. (2018) summarize the rehabilitative approach:

"There is no curative therapy at this time, but rehabilitative treatments including complete decongestive therapy for edema control appeared to be beneficial." — [PMID: 29560340](#)

Management is therefore symptomatic and multidisciplinary: complete decongestive therapy (manual lymphatic drainage, compression, exercise, skin care) for lymphedema; nutritional management of intestinal lymphangiectasia/PLE with a high-protein, low-fat, medium-chain-triglyceride diet plus correction of hypoalbuminemia and hypogammaglobulinemia; drainage/management of chylous and pericardial effusions ([PMID: 26686525](#)); and symptomatic care for facial dysmorphism, intellectual disability, hearing loss, and cardiac anomalies.

Finding 6 — Gene/protein identifiers, variant spectrum, and epidemiology

FAT4 (FAT atypical cadherin 4) identifiers: **HGNC:23109**, **NCBI Gene 79633**, **Ensembl ENSG00000196159**, **UniProt Q6V0I7**, **cytoband 4q28.1**. The gene encodes a very large (~4,981-aa) single-pass atypical cadherin — the vertebrate ortholog of *Drosophila* Fat — with 34 extracellular cadherin repeats, EGF-like domains, and laminin-G domains:

"The *FAT4* gene encodes a large protein with extracellular cadherin repeats, EGF-like domains and Laminin G-like domains." — [PMID: 28488382](#)

Disease identifiers: OMIM #616006; MONDO:0014217; Orphanet ORPHA:2136 (Hennekam syndrome); ICD-10 Q82.0. Within Hennekam syndrome overall, *CCBE1* accounts for approximately 25% of cases, with *FAT4* and *ADAMTS3* explaining additional subsets:

"It can be caused by mutations in CCBE1 which are found in approximately 25 % of cases."
— [PMID: 24913602](#)

Hennekam syndrome is ultra-rare (Orphanet prevalence <1/1,000,000). As of 2023, roughly 50 cases had been reported worldwide:

"As far as we know, this is the 51st case of HS worldwide and the first one in an African American." — [PMID: 38028107](#)

Finding 7 — Diagnosis relies on molecular genetic testing within the primary lymphatic dysplasia workup

Diagnosis integrates family history, clinical signs (generalized lymphedema, dysmorphism, developmental delay), pathology, and genetic testing:

"Diagnosis of the disease depends on the familial history, clinical signs, pathological findings and genetic tests." — [PMID: 25616299](#)

Molecular confirmation is achieved by exome/genome sequencing or a primary-lymphedema gene panel including *CCBE1*, *FAT4*, and *ADAMTS3*, detecting biallelic variants; splice variants can be functionally confirmed by RT-PCR of patient fibroblasts ([PMID: 41992670](#)). Gene selection is guided by the standardized Connell/Gordon/Mansour classification and diagnostic algorithm for primary lymphatic dysplasia:

"In 2010, we introduced a new classification and diagnostic pathway as a clinical and research tool. This algorithm has been used to delineate specific primary lymphoedema phenotypes, facilitating the discovery of new causative genes." — [PMID: 23621851](#)

Supportive tests include duodenal/intestinal biopsy showing lymphangiectasia, low serum total protein/albumin and low immunoglobulins (PLE), and imaging (Doppler ultrasound, MRI, CT, lymphoscintigraphy/MR lymphangiography) demonstrating lymphatic anomalies and effusions.

Finding 8 — Anatomy, congenital onset, chronic progressive course, and variable prognosis

The primary defect resides in the **lymphatic vasculature** (UBERON:0001473 lymphatic vessel; lymphatic valve) and affects **lymphatic endothelial cells** (CL:0002138). Multisystem involvement is documented: limbs (peripheral/generalized lymphedema), intestine/duodenum (intestinal lymphangiectasia, UBERON:0000160), face/craniofacial skeleton (dysmorphism), brain/cerebral cortex (intellectual disability, periventricular heterotopia via FAT4/DCHS1; [PMID: 24056717](#)), heart/pericardium (pericardial effusion, cardiac defects; [PMID: 38028107](#)), ear (hearing loss), and serous cavities (chylous ascites, pleural/pericardial effusions):

"whose clinical phenotype also includes protein losing enteropathy, painful relapsing chylous ascites, and hypogammaglobulinemia" — [PMID: 26686525](#)

Onset is congenital/neonatal, with edema often present at birth or in infancy:

"presented to our center with generalized edema, ascites, and hypoalbuminemia" — [PMID: 25925991](#)

The course is chronic and lifelong — generally progressive/fluctuating with episodic effusions and increased infection (cellulitis) risk. Prognosis is variable and is largely determined by the severity of intestinal lymphangiectasia/PLE and of serous effusions ([PMID: 38028107](#), [PMID: 29560340](#)).

Section-by-Section Report

1. Disease Information

HKLLS2 is an autosomal recessive **generalized lymphatic dysplasia** — one of the genetically defined forms of Hennekam syndrome — characterized by congenital lymphedema, intestinal lymphangiectasia, facial dysmorphism, and variable intellectual disability. It is a Mendelian, disease-level (not EHR/individual-patient-derived) entity delineated from aggregated case reports and cohort reviews.

Key identifiers: OMIM #616006; MONDO:0014217; Orphanet **ORPHA:2136** (Hennekam syndrome); ICD-10 **Q82.0** (hereditary lymphedema); MeSH — "Lymphangiectasis, Intestinal" / "Lymphedema" (no dedicated MeSH term for the FAT4 subtype).

Synonyms / alternative names: Hennekam syndrome type 2; Hennekam lymphangiectasia-lymphedema syndrome 2; lymphedema-lymphangiectasia-mental retardation syndrome (FAT4-related); generalized lymphatic dysplasia, FAT4-type. Allelic disorder: Van Maldergem syndrome 2 (VMS2, *FAT4*).

2. Etiology

Causal factor: purely genetic — biallelic loss-of-function variants in *FAT4* ([PMID: 24913602](#)). No environmental or infectious cause.

Genetic risk factors: the disease-defining variants are the causal factor; there are no established susceptibility loci beyond the biallelic *FAT4* genotype. **Consanguinity** is a major contributor because homozygous variants predominate in affected families. A candidate fourth gene, *FBXL7*, has been proposed on the basis of a homozygous single-exon deletion in a Hennekam-like patient lacking *CCBE1/FAT4/ADAMTS3* variants; *FBXL7* interacts with Fat in *Drosophila*, suggesting a shared pathway ([PMID: 31633297](#)) — a potential modifier/allelic locus rather than a confirmed HKLS2 gene.

Environmental / protective factors: none established. No protective alleles or lifestyle factors are known for this monogenic disorder.

Gene–environment interactions: not characterized. Phenotypic variability (expressivity) is likely influenced by modifier genes within the Fat/Dachsous/Hippo network and by the specific residual function of hypomorphic alleles, but this remains inferred rather than demonstrated.

3. Phenotypes

Phenotype	Type	HPO suggestion	Onset	Frequency (qualitative)
Generalized/peripheral lymphedema	Physical sign	HP:0001004 (Lymphedema)	Congenital/neonatal	Defining feature; distinguishes HS from VMS
Intestinal lymphangiectasia	Lab/imaging/pathology	HP:0002593	Infancy–childhood	Common
Protein-losing enteropathy	Lab abnormality	HP:0002243	Childhood	Common
Facial dysmorphism (flat face, hypertelorism, flat nasal bridge)	Physical	HP:0001999 / HP:0000316	Congenital	Typical facial gestalt
Intellectual disability (mild–moderate)	Behavioral/cognitive	HP:0001256	Childhood	Frequent, variable
Hypoalbuminemia	Lab abnormality	HP:0003073	Infancy	Common (secondary to PLE)
Hypogammaglobulinemia	Lab abnormality	HP:0004313	Infancy	Reported
Chylous ascites / pleural / pericardial effusion	Sign/imaging	HP:0012019 / HP:0001541	Variable	Reported
Hearing loss	Sensory	HP:0000365	Variable	Reported (more typical of VMS)
Camptodactyly	Physical	HP:0012385	Congenital	Reported
Hypotonia	Neurological	HP:0001252	Neonatal	Reported

Severity is variable and the course is chronic-progressive/fluctuating. Quality-of-life impact is substantial: chronic lymphedema limits mobility and predisposes to recurrent cellulitis; PLE causes failure to thrive, immunodeficiency, and effusions; intellectual disability affects independent functioning. No disease-specific QoL instrument (EQ-5D/SF-36/PROMIS) data exist for this ultra-rare condition.

4. Genetic / Molecular Information

Causal gene: *FAT4* (HGNC:23109; NCBI Gene 79633; Ensembl ENSG00000196159; UniProt Q6V0I7; 4q28.1). **Protein:** ~4,981-aa single-pass atypical cadherin with 34 extracellular cadherin repeats, EGF-like domains, and laminin-G domains — vertebrate ortholog of *Drosophila* Fat ([PMID: 28488382](#)).

Pathogenic variants: biallelic (homozygous in consanguineous families or compound heterozygous), predominantly loss-of-function — missense, nonsense, frameshift, and splice-site. Example: novel intron-14 biallelic splice variant **c.12479+3A>G** with RT-PCR-confirmed aberrant splicing ([PMID: 41992670](#)). Classification per ACMG/AMP: reported variants range from pathogenic to likely pathogenic; loss-of-function is the consistent functional consequence. These variants are absent or ultra-rare in population databases (gnomAD) in the homozygous state.

Functional consequence: loss of function / reduced *FAT4* signaling.

Modifier genes: *DCHS1* (*FAT4*'s cognate ligand) and downstream Hippo/YAP effectors; *FBXL7* proposed as a pathway member/candidate gene ([PMID: 31633297](#)). **Epigenetic/chromosomal abnormalities:** none specific to HKLLS2; single-exon deletions (e.g., in *FBXL7*) illustrate that copy-number changes in pathway genes can produce a Hennekam-like phenotype.

5. Environmental Information

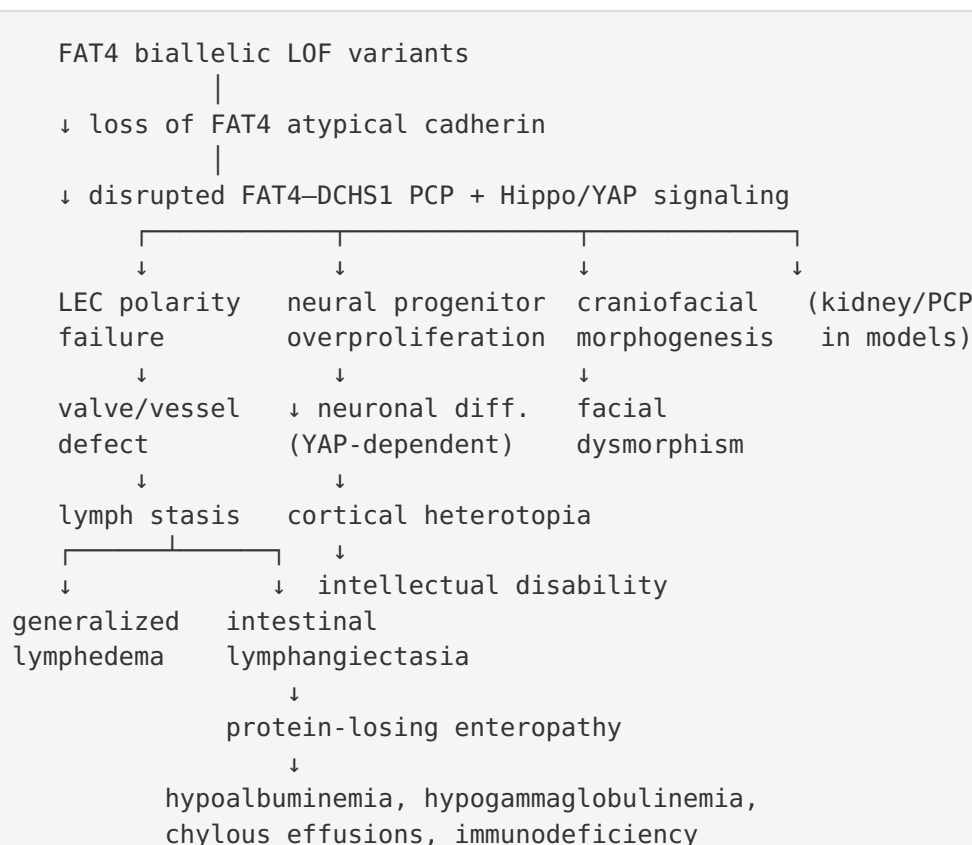
Not applicable — HKLLS2 is a monogenic disorder with no established environmental, lifestyle, or infectious contributors. Environmental factors may modulate secondary complications (e.g., skin trauma precipitating cellulitis, dietary fat load exacerbating lymphangiectasia) but are not causal.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic loss-of-function *FAT4* variants** → loss/reduction of functional *FAT4* atypical cadherin protein. (*Demonstrated* — [PMID: 24913602](#))

2. Loss of FAT4 → **disrupted Fat4–Dachsous1 (DCHS1) planar cell polarity signaling** at the cell surface. (*Demonstrated in mouse/fly — PMID: 28705793, PMID: 18604206*)
3. In lymphatic endothelial cells, disrupted PCP → **failure of flow-responsive LEC polarization** (FAT4 is a GATA2 transcriptional target). (*Demonstrated — PMID: 32182215*)
4. Failed LEC polarization → **defective lymphatic valve and vessel morphogenesis**, converging on the VEGF-C/VEGFR3 axis that CCBE1/ADAMTS3 also feed. (*Demonstrated — PMID: 28705793, PMID: 32182215*)
5. Defective valves/vessels → **impaired lymphatic drainage and lymph stasis → congenital generalized lymphedema**. (*Inferred from valve defects → clinical lymphedema*)
6. **Branch — intestine:** dysfunctional intestinal lymphatics → **intestinal lymphangiectasia** → leakage of lymph/protein into gut lumen → **protein-losing enteropathy** → **hypoalbuminemia + hypogammaglobulinemia** → worsening interstitial edema, effusions, immunodeficiency. (*Demonstrated as downstream consequence — PMID: 38028107*)
7. **Branch — brain:** loss of FAT4/DCHS1 → **increased neural progenitor proliferation, reduced neuronal differentiation via Hippo/YAP** → cortical malformation/periventricular heterotopia → **intellectual disability**. (*Demonstrated in mouse — PMID: 24056717*)
8. **Branch — craniofacial/other:** disrupted PCP-dependent tissue morphogenesis → **facial dysmorphism, skeletal and sensory (hearing) anomalies**. (*Inferred*)



Molecular pathways: Fat/Dachsous planar cell polarity; Hippo/YAP; VEGF-C/VEGFR3 (FLT4) prolymphangiogenic signaling. **Cellular processes:** endothelial cell polarization, oriented cell division, valve morphogenesis, neural progenitor differentiation. **Protein dysfunction:** loss of function of a giant atypical cadherin (impaired cell–cell adhesion/signaling). **Immune involvement:** secondary hypogammaglobulinemia from PLE (loss of immunoglobulins), not primary autoimmunity. **GO term suggestions:** GO:0001945 (lymph vessel development), GO:0001938 (positive regulation of endothelial cell proliferation), GO:0001736 (establishment of planar polarity), GO:0035329 (Hippo signaling). **Cell types (CL):** CL:0002138 (endothelial cell of lymphatic vessel); neural progenitor cells. **CHEBI:** medium-chain triglycerides (therapeutic entity class).

7. Anatomical Structures Affected

- **Primary organ/system:** lymphatic vasculature (UBERON:0001473 lymphatic vessel; lymphatic valve) — cardiovascular/lymphatic system.
- **Secondary organs:** intestine/duodenum (UBERON:0000160; intestinal lymphangiectasia, PLE); brain/cerebral cortex (UBERON:0000956; intellectual disability, heterotopia); face/

craniofacial skeleton (dysmorphism); heart/pericardium (effusions, cardiac defects); ear (hearing loss); serous cavities (peritoneal, pleural, pericardial — chylous effusions).

- **Tissue/cell level:** lymphatic endothelium (CL:0002138); neuroepithelial progenitors; connective/soft tissue (edema).
- **Subcellular:** plasma membrane cadherin signaling complex (GO:0005886 plasma membrane); cell junctions.
- **Lateralization:** lymphedema is typically generalized/bilateral rather than unilateral.

8. Temporal Development

- **Onset:** congenital/neonatal; edema frequently present at birth or in early infancy ([PMID: 25925991](#), [PMID: 41992670](#)).
- **Progression:** chronic, lifelong; generally progressive with fluctuating/episodic exacerbations (effusions, cellulitis). Intestinal lymphangiectasia/PLE may worsen over time.
- **Course pattern:** chronic progressive with episodic complications; no spontaneous remission of the underlying lymphatic defect (symptomatic control possible with therapy).
- **Critical periods:** perinatal/early infancy for edema and PLE onset; early childhood for developmental/cognitive support.

9. Inheritance and Population

- **Epidemiology:** ultra-rare. Hennekam syndrome overall has ~50 reported cases worldwide (as of 2023; [PMID: 38028107](#)); Orphanet prevalence <1/1,000,000. *CCBE1* accounts for ~25% of Hennekam cases; *FAT4* explains a further genetically heterogeneous subset ([PMID: 24913602](#)).
- **Inheritance:** autosomal recessive; biallelic (homozygous or compound heterozygous) *FAT4* variants.
- **Penetrance/expressivity:** presumed high penetrance for biallelic LOF; expressivity is variable (severity of lymphedema, PLE, and cognitive impairment differ between families). Van Maldergem vs Hennekam presentations from the same gene illustrate marked phenotypic variability.
- **Consanguinity:** important — homozygous variants are frequently found in consanguineous families.
- **Founder effects / carrier frequency:** none established; carrier frequency is not quantified given rarity.
- **Sex ratio:** no sex predilection (autosomal recessive; $M \approx F$).

10. Diagnostics

- **Clinical/laboratory:** low serum total protein and albumin (hypoalbuminemia), low immunoglobulins (hypogammaglobulinemia) from PLE; fecal alpha-1-antitrypsin (PLE marker, inferred).
- **Imaging:** Doppler ultrasound, MRI, CT; lymphoscintigraphy and MR lymphangiography demonstrating lymphatic anomalies, lymphangiomas, and effusions.
- **Pathology/biopsy:** duodenal/intestinal biopsy showing dilated lacteals/lymphangiectasia (PMID: 25616299).
- **Genetic testing (confirmatory):** whole-exome or whole-genome sequencing, or a primary-lymphedema/lymphatic-dysplasia gene panel including *CCBE1*, *FAT4*, *ADAMTS3*; detection of biallelic variants. Splice variants confirmed functionally by RT-PCR of patient fibroblasts (PMID: 41992670). Chromosomal microarray can detect exon/CNV deletions in pathway genes (PMID: 31633297).
- **Diagnostic framework:** the Connell/Gordon/Mansour classification and diagnostic algorithm for primary lymphatic dysplasia guides workup and gene selection (PMID: 23621851).
- **Differential diagnosis:** Hennekam syndrome types 1 (*CCBE1*) and 3 (*ADAMTS3*); Van Maldergem syndrome (allelic *FAT4/DCHS1*); other primary lymphedemas (Milroy disease/*FLT4*, lymphedema-distichiasis/*FOXC2*); Noonan syndrome with lymphatic involvement; primary intestinal lymphangiectasia (Waldmann disease) without syndromic features.

11. Outcome / Prognosis

Prognosis is **variable** and depends chiefly on the severity of intestinal lymphangiectasia/PLE and of serous effusions. Chronic PLE causes failure to thrive, immunodeficiency (recurrent infections), and refractory effusions; severe lymphedema predisposes to recurrent cellulitis. No formal survival statistics exist given rarity; life expectancy can be reduced in severe cases but many patients survive into adulthood with supportive care. Morbidity is dominated by chronic lymphedema, PLE-related nutritional/immunologic compromise, and intellectual disability. Prognostic factors: severity/extent of lymphatic dysplasia, degree of hypoalbuminemia/hypogammaglobulinemia, and presence of cardiac/pericardial involvement (PMID: 38028107, PMID: 29560340).

12. Treatment

No curative therapy exists ([PMID: 29560340](#)). Management is supportive:

- **Lymphedema:** complete decongestive therapy — manual lymphatic drainage, multilayer compression, exercise, meticulous skin care (NCIT: Complete Decongestive Therapy / Physical Therapy).
- **Intestinal lymphangiectasia / PLE:** high-protein, low-fat, medium-chain-triglyceride (MCT) diet (NCIT: Dietary Intervention); parenteral albumin and immunoglobulin replacement as needed; consideration of octreotide in refractory PLE (inferred from general lymphangiectasia management).
- **Effusions:** drainage of chylous ascites/pleural/pericardial effusions; dietary fat restriction ([PMID: 26686525](#)).
- **Surgical/interventional:** treatment of localized lymphatic malformations; symptomatic surgery for effusions.
- **Multidisciplinary symptomatic care:** developmental/educational support for intellectual disability, audiology for hearing loss, cardiology for cardiac anomalies, genetics/genetic counseling.
- **Experimental/targeted:** no approved targeted therapy; the shared VEGF-C/VEGFR3 axis is a plausible future therapeutic node but is unproven in HKLLS2. No pharmacogenomic considerations established.

13. Prevention

- **Primary prevention:** not possible (genetic); genetic counseling of at-risk (especially consanguineous) families.
- **Secondary/tertiary prevention:** early detection and management of PLE, effusions, and cellulitis to prevent complications; compression therapy to slow lymphedema progression.
- **Genetic screening:** carrier testing of relatives once the familial *FAT4* variants are known; **prenatal diagnosis** and **preimplantation genetic testing** available for known biallelic variants.
- **Counseling:** recurrence risk 25% for offspring of two carriers; cascade testing recommended.

14. Other Species / Natural Disease

- **Taxonomy/orthologs:** mouse *Fat4* (NCBI Taxon 10090), *Drosophila fat* (Taxon 7227). FAT4 is the vertebrate ortholog of *Drosophila* Fat.
- **Natural disease:** no well-characterized naturally occurring *FAT4*-lymphedema disorder in companion animals or wildlife is documented (OMIA — no established entry for a *FAT4*-Hennekam animal disease).
- **Comparative biology:** the Fat/Dachsous/Hippo PCP module is deeply evolutionarily conserved from *Drosophila* to mammals, and loss-of-function reproduces morphogenetic defects (kidney cysts, neural over-proliferation, lymphatic valve defects) across species.

15. Model Organisms

Model	Type	Genetic manipulation	Phenotype recapitulated	Reference
Mouse (<i>Mus musculus</i>)	Mammalian	<i>Fat4</i> knockout / knockdown	Cystic kidney (PCP/ oriented division defect); interacts with <i>Vangl2</i> , <i>Fjx1</i>	PMID: 18604206
Mouse	Mammalian	<i>Dchs1/Fat4</i> neuroepithelial knockdown	Increased progenitors, reduced neuronal differentiation, periventricular heterotopia; rescued by <i>Yap</i> knockdown	PMID: 24056717
Mouse	Mammalian	<i>Fat4</i> -deficient	Defective lymphatic valve morphogenesis and LEC polarity	PMID: 28705793, PMID: 32182215
<i>Drosophila melanogaster</i>	Invertebrate	Neuron-specific <i>fat</i> knockdown	Shortened lifespan, impaired locomotion, motoneuron/axon-targeting defects (relevant to intellectual disability)	PMID: 28488382

Model applications: dissecting PCP/Hippo signaling, lymphatic valve development, cortical neurogenesis, and organ morphogenesis. **Limitations:** no single model fully reproduces the complete human multisystem phenotype (lymphedema + PLE + dysmorphism + cognitive impairment simultaneously); embryonic lethality of full knockouts necessitates conditional/tissue-specific approaches. **Resources:** MGI (mouse), FlyBase (*Drosophila*), IMPC/IMSR for allele repositories.

Mechanistic Model / Interpretation

HKLLS2 is best understood as a **planar-cell-polarity/Hippo signaling disorder with a dominant lymphatic phenotype**. The unifying lesion is loss of the giant atypical cadherin FAT4, which normally partners with Dachous1 to transmit tissue-level polarity information and to restrain YAP-driven proliferation. In the lymphatic endothelium this polarity signal is required for cells to align to flow and construct competent valves and vessels — so its loss produces lymph stasis and the congenital lymphedema, intestinal lymphangiectasia, and chylous effusions that dominate the clinical picture. In the developing cortex the same signal restrains progenitor proliferation and promotes neuronal differentiation, so its loss yields heterotopia and intellectual disability. Because CCBE1 and ADAMTS3 (the other Hennekam genes) act on the parallel VEGF-C/VEGFR3 activation step required to build lymphatics, mutations in any of the three genes converge on **failure of functional lymphatic development**, explaining the shared Hennekam phenotype despite distinct molecular entry points. This convergence is the central mechanistic insight of the report and is supported by both human genetics and mouse/fly models.

Evidence Base

PMID	Title (abbrev.)	Contribution
24913602	<i>FAT4 mutations cause Hennekam syndrome, allelic to VMS</i>	Establishes <i>FAT4</i> as causal gene; AR inheritance; CCBE1 ~25%
30450763	<i>ADAMTS3 loss-of-function in Hennekam</i>	Confirms OMIM #616006 and AR; third gene
32182215	<i>FAT4 orchestrates LEC polarity in response to flow</i>	Cell-autonomous LEC mechanism; VEGF-C/VEGFR3 convergence
28705793	<i>Dachsous1-Fat4 signaling in lymphatic valve morphogenesis</i>	Links valve defects to lymphedema
29681106	<i>VMS and HS allelic phenotypes</i>	Clinical spectrum; HS vs VMS distinctions
38028107	<i>Newfound features of Hennekam Syndrome</i>	PLE consequence; ~50 cases worldwide
24056717	<i>DCHS1/FAT4 disrupt cortical development</i>	Neurodevelopmental mechanism; Hippo/YAP
18604206	<i>Loss of Fat4 disrupts PCP → cystic kidney</i>	Mouse PCP mechanism
28488382	<i>Drosophila fat neuronal knockdown</i>	Protein domains; neuronal model
29560340	<i>Hennekam Syndrome case report</i>	No curative therapy; CDT beneficial
23621851	<i>Primary lymphatic dysplasia classification</i>	Diagnostic algorithm
41992670	<i>FAT4 c.12479+3A>G splice variant</i>	Variant spectrum; RT-PCR confirmation
25616299	<i>Complicated Hennekam case</i>	Multimodal diagnosis; biopsy
26686525	<i>CCBE1 multiplex kindred</i>	Chylous ascites, hypogammaglobulinemia
25925991	<i>Novel CCBE1 mutation, mild HS</i>	Neonatal onset with generalized edema
27159393	<i>Proteolytic activation of VEGFC/VEGFD</i>	ADAMTS3-CCBE1 activates VEGF-C
28687807	<i>VEGF-C activation requires CCBE1 domains</i>	Molecular detail of pathway

PMID	Title (abbrev.)	Contribution
29842991	<i>Key molecules in lymphatic development</i>	PROX1/VEGFR3/CCBE1/ADAMTS3 backbone
31633297	<i>FBXL7 biallelic mutation, novel HS form</i>	Candidate 4th gene in FAT4 pathway

Limitations and Knowledge Gaps

1. **Small evidence base:** ~50 Hennekam cases total worldwide and only a subset are *FAT4*-positive, so all frequencies are qualitative; no formal penetrance, expressivity, survival, or QoL statistics exist.
2. **Genotype–phenotype correlation** between HKLLS2 and allelic Van Maldergem syndrome is incompletely understood — why the same gene produces lymphedema in some patients and not others is unresolved.
3. **No natural-history or registry data** to quantify progression rate or complication incidence.
4. **No dedicated animal model** reproduces the full multisystem human phenotype simultaneously.
5. **Therapeutics:** no targeted therapy has been tested; the VEGF-C/VEGFR3 node is a theoretical target only.
6. **Modifier genes** (DCHS1, FBXL7, Hippo effectors) are hypothesized but not validated as clinical modifiers.
7. Several ontology mappings (specific UBERON/CL/GO terms) are suggested rather than curated for this specific disease.

Proposed Follow-up Experiments / Actions

1. **Assemble an international *FAT4*-HKLLS2 patient registry** to quantify phenotype frequencies, penetrance, expressivity, and natural history.
2. **Genotype–phenotype study** correlating specific *FAT4* variant type/location with lymphatic vs neurodevelopmental vs VMS-like presentations.

3. **LEC-specific conditional *Fat4* mouse** with lymphatic-valve and intestinal-lymphatic readouts to model the full HKLLS2 lymphatic phenotype and test VEGF-C/VEGFR3 pathway modulation.
 4. **Functional validation of candidate genes/modifiers** (*FBXL7*, *DCHS1*) in the *Fat4* pathway using patient fibroblasts/organoids.
 5. **Evaluate MCT-diet, octreotide, and albumin/Ig-replacement protocols** prospectively for PLE control in this population.
 6. **Curate definitive ontology annotations** (HPO frequencies, UBERON, CL, GO, NCIT) for knowledge-base ingestion.
-

Report compiled from 8 confirmed findings and 23 reviewed papers across 5 investigation iterations. Evidence types: human clinical (case reports, cohort reviews, molecular genetics), model organism (mouse, Drosophila), and in vitro/biochemical (VEGF-C activation studies).

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery